PAGE 13 – BUG AND DEBUGGING LOG

Record major bugs encountered during system development and debugging:

No.	Error / Bug Encountered	Root Cause	Solution Applied	Result
1	Compiler error: “redefinition of struct Book”	Book was defined in both EntityModel.h and AuditStackModule.h, and main.cpp includes both.	Deleted the duplicate struct from the stack header and added #include "EntityModel.h".	Error gone. Project uses one shared Book.
2	Errors: “redefinition of ActionType, AuditLog, AuditStackModule ” and push	The stack .cpp had a full copy of the code pasted at the top, including its own structs and a test main(), on top of the header.	Replaced the .cpp with a clean version holding only the member-function implementations.	Build errors cleared.
3	Stack files not found or duplicated in the project	Header and source names didn’t match the #include lines (StackEngine vs AuditStackModule), and two file pairs were in the project.	Kept one file pair and made every #include use that exact name.	Include errors resolved.
4	Undo of a book insertion couldn’t find which book to remove.	The stack’s pop() only returns the previous state, which is empty for an insertion, so the ISBN was lost.	Stored a placeholder book holding only the ISBN when an insertion is logged.	Undo of an insertion can now identify the right book.
5	“Not declared” error in SearchSortEngine. cpp	Missing #include	Added the missing #include at the top of the SearchSortEngine File.	Error cleared and the file compiled.

DEBUGGING REFLECTION

Most difficult bug encountered during development: The “redefinition of struct Book” error in the audit stack module. It was hard because it caused a long chain of other errors at once (ActionType, AuditLog, AuditStackModule, and the push/pop functions were all reported as redefined). The real problem was small, but the messages made it look like many separate bugs.

How was it identified and resolved?

- When I ran the code, it showed repeated errors related to the Stack and Entity files. After checking the Build messages, I found that the Book struct was defined in both EntityModel.h and the stack header, which caused a conflict. The stack.cpp file also contained duplicate code. I fixed this by removing the duplicate definition, including EntityModel.h in the stack header, and cleaning up the .cpp file. I also checked that the file names and #include statements matched. After rebuilding the project, the errors were gone. From this, I learned to check the first compiler error and make sure the same structure is not defined multiple times.